

Stat 3113: Lesson 10

Random Variables



- It frequently occurs that in performing an experiment, we are mainly interested in some functions of the outcome as opposed to the outcome itself.
- The value of a Random variable is determined by the outcome of the experiment. $X(\omega)$, $\omega \in S$.

- Recall the definition of variable (Lesson 1) [characteristic changing over time and/or for different individual under consideration]

Definition: A **random variable (rv)** is a function which maps each outcome in the sample space to a real number.

$$\begin{aligned} X: S &\longrightarrow \mathbb{R} \\ \omega &\longmapsto X(\omega) \end{aligned}$$

Typical Notation:

- Random variables:** Capital letters like X, Y, Z , etc.
- Values of the random variable:** Lower case letters like x, y, z , etc.
- The statement means the random variable X takes on the value x .

Examples:

a. Toss a coin twice. Let $X = \#$ of heads observed. What are the possible values of X ?

$$S = \Omega = \{HH, TH, HT, TT\} \Rightarrow X(\omega) \in \{0, 1, 2\}$$

- b. Roll a pair of dice. Let X = maximum value observed on one die. Let Y = sum of the two numbers observed. Let $Z = \begin{cases} 1, & \text{if pair match} \\ 0, & \text{if otherwise} \end{cases}$

What are the possible values of X , Y , and Z ?

↳ Also called the domain/support of X .

- $S = \{ (w_1, w_2) \mid w_1, w_2 \in \{1, 6\} \}$ ← Set builder notation from Lesson 6
- $X(w) \in \{1, 2, 3, 4, 5, 6\}$, $\forall w \in S$ ($\forall :=$ for all)
- $Y(S) = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\} = \{2, 12\}$
- $Z(S) = \{0, 1\}$ → Z is also called a dummy/binary variable in general.
↓ when its possible values are 0 and 1.

Z is called a **Bernoulli random variable**, a random variable whose only possible values are 0 and 1.

- Discrete variable $\rightarrow$ It can assume only a finite or countable number of values.

$\hookrightarrow$ Countable $\rightarrow$ its elements can be put in a one-to-one mapping with the natural numbers.

Ex: A growing population $\rightarrow \mathbb{N}$.

Ex:- Number of defects in products on a production line.
- Number of student in this class! 36!

- Continuous variable $\rightarrow$ It can assume the infinitely many values corresponding to the points on a line interval.

Ex:- Time to failure of a Computer component!

- There is an infinite number of possible values between any two given temperatures.

Discrete and Continuous Random Variables

- A **discrete** random variable is a random variable whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, a second element, and so on ("countably" infinite).
- A random variable is **continuous** if *both* of the following apply:
 1. Its set of possible values consists either of all numbers in a single interval on the number line or all numbers in a disjoint union of such intervals (ex: $[0, 10] \cup [20, 30]$).
 2. No possible value of the variable has positive probability, that is, $P(X = c) = 0$ for any possible value c .

Example: Discrete or Continuous rv?

- a. $X = 1, 2, 3, 4, 5, 6$ $\longrightarrow$ Discrete (Finite)
- b. $Y = 1, 2, 3, \dots$ $\longrightarrow$ Discrete (Countable)
- c. $X \geq 0$ $\longrightarrow$ Continuous

- Next Up?

- ↳ We may assign probabilities to the possible values of a random variable.

- Ex: The experiment consisting of rolling a pair of dice.

- $$IP(\{(1,2), (2,1)\}) = \frac{2}{36}.$$

- $X =$ Sum of two fair dice.

- $$IP(\{X=3\}) = IP(\{(1,2), (2,1)\}) = \frac{2}{36}.$$

$X = \text{Discrete.}$

↳ Probability distribution

↳ Complete representation of our uncertainty about events relating to X .



Domain/Support
 $X(s)/X(\omega)$

Functions which assign probabilities to the possible values of X in $X(s)$.



• $\text{pmf} = f(x) = \mathbb{P}(X=x)$

• $\text{CDF} = F(x) = \mathbb{P}(X \leq x)$

$$= \sum_{x_i \leq x} \mathbb{P}(X=x_i)$$

Reading

Chapter 3 – Section 3.3

Next Topic



Discrete Random Variables



Chapter 3 - Section 3.3